

Throughput-Optimized Multi-Rate LDPC Codes for Satellite Communication Systems With Restricted Decoder Complexity

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Abstract—Multi-rate low-density parity-check (LDPC) codes are widely used in satellite communication (SatCom) systems due to their outstanding performance and adaptable code rates. However, their hardware decoder implementations typically require high hardware resources and many decoding iterations, which pose challenges for high-throughput applications in resource-constrained, high-data-rate SatCom scenarios. To this end, the design of throughput-optimized multi-rate LDPC codes is investigated, aiming to tackle the decoder implementation problem from the perspective of practical code design. Specifically, a multi-rate weighted normalized throughput metric is introduced, which considers two key factors, i.e., the number of non-zero elements in the parity-check matrix and the required decoding iterations. Then, the throughput-optimized multi-rate code design problem is formulated to maximize the proposed metric under given decoding threshold constraints, which is later solved using a protomatrix search method based on the differential evolution (DE) algorithm. Finally, three multi-rate LDPC codes are optimized for scenes with different code rate usage frequencies. Their performance is extensively evaluated both theoretically and numerically. Results demonstrate that the proposed codes outperform other similar LDPC codes in the literature, showing a throughput increase of 20% to 30% under the same hardware resource consumption.

Index Terms—LDPC codes, code design, throughput optimization, resource-constrained systems, decoding complexity, satellite communication.

I. INTRODUCTION

A. Motivations

WITH the growing demand for high-speed, stable and seamless communication, high-data-rate satellite communication (SatCom) systems have aroused increasing attention in recent years. These systems are expected to complement

terrestrial communication networks by providing high-speed, low-latency services for a wide range of applications, including satellite broadband communication [1], [2], [3], [4], high-speed laser communication [5], [6], and high-resolution remote sensing data transmission [7], [8], [9]. Among these, LEO SatCom systems, in the form of mega-constellations such as Starlink, OneWeb, and Project Kuiper, have emerged as particularly promising technologies for providing broadband services, owing to their short propagation delay, low path loss and cost-effectiveness [10], [11]. Meanwhile, MEO satellite constellations offer the benefits of larger coverage areas and enhanced link stability, playing a critical role in increasing capacity and data rates. To ensure reliable data transmission, channel coding techniques are essential in SatCom systems. Modern channel codes with high coding gains, such as low-density parity-check (LDPC) codes [12] and Turbo codes [13], have been extensively studied and widely implemented in traditional SatCom systems. Among these, multi-rate LDPC codes [14], [15], [16] stand out for their capacity-approaching decoding performance, inherent parallelizability, and flexible code rates. These features make multi-rate LDPC codes highly competitive candidates for future SatCom systems, and they have been the focus of extensive research in recent studies [14], [15], [16].

Nevertheless, in the context of high-data-rate SatCom systems, the feasibility and effectiveness of existing multi-rate LDPC coding schemes must be re-evaluated. A critical consideration is hardware consumption. To reduce deployment costs, commercial satellites have increasingly been miniaturized in recent years [17], resulting in limited onboard hardware resources. However, LDPC decoders, which typically consume far more hardware resources than their encoder counterparts, require substantial computational resources. As a result, decoding complexity becomes a primary concern and the central focus of this work under these constraints. Another key challenge is achieving sufficient data throughput. As SatCom systems are expected to support high-throughput services [18], [19], LDPC codes, which rely on iterative decoding, typically require tens of iterations to achieve the optimal coding gain, thereby limiting data throughput. Although reducing the number of iterations can improve throughput, it often comes at the cost of degraded decoding performance. Therefore, it is essential to design LDPC codes that maintain high decoding performance while operating with a reduced number of iterations. In addition to these general

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challenges, the support for multiple LDPC code rates introduces further design difficulties. Decoding multiple code rates requires additional hardware to support flexible code structures and scheduling, leading to increased resource consumption on constrained satellite platforms. Furthermore, optimizing performance and resource usage across multiple code rates simultaneously is nontrivial, as it is challenging to design a single parity-check matrix that performs well across all rates and aligns with the requirements of the intended application scenarios. Addressing these challenges underscores the need to explore hardware-efficient, high-throughput multi-rate LDPC coding solutions that can operate effectively within the constraints of both hardware resources and throughput requirements.

Most existing studies focus on developing low-complexity decoding algorithms or decoder architectures to achieve high-throughput LDPC decoders with minimal hardware resource consumption. However, recent research suggests that further optimization is possible at the code design stage [20], [21]. By establishing the relationship between resource consumption, decoding throughput, and the code matrix, it becomes possible to design LDPC codes that achieve both low hardware complexity and excellent performance with a small number of iterations. Moreover, these optimized codes can be effectively combined with low-complexity algorithms and decoder architectures to significantly reduce resource consumption while enhancing data throughput. Given the specific requirements of SatCom systems for channel coding schemes, this paper focuses on the design of high-throughput multi-rate LDPC codes with restricted decoder complexity to meet the hardware resource limitations and data throughput demands of high-data-rate SatCom systems.

B. Related Works

LDPC codes are commonly characterized by their parity-check matrices (PCM), and the design of these matrices plays a central role in code optimization. Traditionally, most research has focused on threshold-optimized LDPC codes, where the objective is to minimize the decoding threshold to improve coding gain under iterative decoding. In this work, we focus on designing LDPC codes that minimize resource consumption and maximize decoding throughput while ensuring specific decoding performance requirements. Two key factors are critical in designing such codes, i.e., the number of non-zero entries in the LDPC parity-check matrix, which directly impacts hardware resource usage, and the number of decoding iterations, which affects decoding throughput.

Recent studies have started to explore LDPC code design from these perspectives. Their results have been shown to require fewer hardware resources or exhibit higher decoding throughput compared to traditional threshold-optimized LDPC codes while maintaining comparable decoding performance. For single-rate LDPC codes, Smith et al. [20] defined the product of the above two parameters as the decoding complexity metric to search for complexity-optimized irregular LDPC codes. Specifically, their approach fine-tunes existing threshold-optimized LDPC codes to derive those with lower decoding complexity. In [21], a low-complexity protograph LDPC (P-LDPC) code with a rate of

$1/2$ was constructed by manually limiting the sum of elements in the protomatrix and restricting the number of iterations to a small value. In [22], a convex optimization framework was formulated to design LDPC codes for the binary erasure channel (BEC) with fast convergence, minimizing the number of decoding iterations required to achieve a target residual erasure probability. For non-binary LDPC codes, [25], [26] focused on performance-complexity tradeoff by optimizing degree distributions to accelerate convergence, achieving approximately 30% reduction in decoding iterations compared to threshold-optimized designs. Furthermore, [27] investigated the use of multilevel coding (MLC) and bit-interleaved coded modulation (BICM) schemes in conjunction with complexity-optimized LDPC inner codes, demonstrating near-capacity performance with manageable decoding complexity. For multi-rate codes, Barakatain et al. [28] recently proposed a rate- and channel-configurable low-complexity concatenated coding scheme using complexity-optimized irregular LDPC codewords as the inner code. Additionally, a delay-limited multi-rate P-LDPC ensemble was designed by Nguyen et al. [24] with small decoding iterations for each code rate using a lengthening method. However, the fast decoding convergence comes at the cost of increased column weight in the parity-check matrix, leading to significantly higher hardware resource usage.

C. Contributions

Although previous studies have made significant progress in optimizing certain aspects of LDPC code construction, few have considered both resource consumption and decoding throughput, which could lead to higher throughput under limited resource consumption. Additionally, most of the existing research focuses primarily on single-rate codes, with limited analysis and optimization of the hardware complexity and decoding throughput for multi-rate LDPC codes. To the best of our knowledge, no research has specifically addressed the optimization and construction of multi-rate LDPC codes for high-data-rate SatCom scenarios with resource constraints.

In response to these limitations, this study focuses on designing throughput-optimized multi-rate LDPC codes that minimize hardware resources usage while maximizing decoding throughput. We aim to optimize normalized throughput, i.e., achievable throughput per unit of resource consumption, for multi-rate LDPC codes. A summary comparison of representative existing works and the proposed approach is provided in Table I, which highlights their differences in code structure, design objectives, and the consideration of decoding throughput and resource usage. A multi-rate weighted normalized decoding throughput metric is proposed, which considers both the number of non-zero entries in the entire parity-check matrix and the decoding iterations required for each code rate. Unlike the conventional rate-by-rate extension method for constructing multi-rate LDPC codes, the proposed search method involves a comprehensive search of the entire parity-check matrix to ensure maximum weighted normalized decoding throughput. To tackle the challenges posed by the large search space, a protomatrix search method based on the differential evolution (DE)

TABLE I
COMPARISON BETWEEN THIS WORK AND REPRESENTATIVE LDPC CODE DESIGN APPROACHES

Work	Code Type	Code Rates	Design Target	Resource Aware	Throughput Aware
Smith et al. [20]	Irregular LDPC	Single-rate	Decoding complexity minimization	✓	✓
Jamali et al. [22]	Irregular LDPC	Single-rate	Decoding iterations minimization	✗	✓
Tang et al. [21]	Protograph LDPC	Single-rate	Threshold optimization with constrained complexity	✓	✓
AR4JA [23]	Protograph LDPC	Multi-rate	Threshold optimization for large iterations	✗	✗
Nguyen et al. [24]	Protograph LDPC	Multi-rate	Threshold optimization for low decoding iterations	✗	✓
This work	Protograph LDPC	Multi-rate	Multi-rate weighted normalized throughput maximization	✓	✓

algorithm [29] is proposed. Analytical and simulation results show that the proposed throughput-optimized multi-rate LDPC codes enhance the decoding throughput by approximately 20% to 30% compared to existing multi-rate LDPC codes, while maintaining similar decoding performance. The primary contributions of this work are outlined as follows:

- 1) We propose a multi-rate weighted normalized throughput metric for designing throughput-optimized multi-rate LDPC codes, which effectively reflects the achievable throughput of an LDPC decoder per unit of hardware resource consumption.
- 2) Based on the proposed metric, we formulate a constrained optimization problem for throughput-optimized LDPC code design. A practical search method based on the differential evolution algorithm is also presented to find optimal solutions efficiently.
- 3) Three multi-rate LDPC codes for scenes with different code rate usage frequencies are constructed using the proposed code design approach. Extensive analyses and comparisons with respect to decoding performance and throughput are presented, demonstrating that the proposed throughput-optimized multi-rate LDPC codes achieve higher decoding throughput in resource-constrained systems without significant performance loss.

The remainder of this paper is organized as follows. In Section II we provide a detailed analysis of the resource utilization and throughput of LDPC decoders, and present the proposed multi-rate weighted normalized decoding throughput metric. Section III formulates the constrained optimization problem and introduces the protomatrix search method. Section IV gives code design examples and presents theoretical analysis and performance comparisons. Section V presents the numerical results and comparisons based on Monte Carlo simulations. Section VI concludes the paper.

II. THROUGHPUT-OPTIMIZED LDPC CODES: KEY FACTORS AND WEIGHTED METRIC

In this section, we first analyze key factors involved in the design of throughput-optimized LDPC codes with limited decoder complexity and present the normalized throughput metric for single-rate LDPC codes. Additionally, for multi-rate LDPC codes, a multi-rate weighted normalized throughput metric is introduced and will be used for code optimization and performance evaluation.

A. Key Factors in Throughput-Optimized LDPC Design

In most LDPC coding schemes, achieving good decoding performance requires a decoder with relatively high hardware resource usage and multiple decoding iterations. A typical LDPC decoder [30], [31] demands substantial layout and routing resources to connect check node units (CNUs) and variable node units (VNUs), as well as considerable random access memory (RAM) resources to store extrinsic information during iterative decoding. Furthermore, they often require numerous iterations to achieve the desired decoding performance, which limits the decoder's throughput and may result in high power consumption [32]. Rather than addressing these challenges by designing elaborate decoding algorithms or decoder architectures, this paper focuses on optimizing complexity and throughput at the initial code design stage, and proposes a class of LDPC codes, referred to as throughput-optimized LDPC codes, that aim to maximize throughput per unit of resource consumption.

In order to design throughput-optimized LDPC codes, as illustrated in Fig. 1, two key factors must be considered in the code construction stage, namely the number of non-zero elements in the parity-check matrix $\mathbf{H}$ as shown in Fig. 1(a) (denoted as E) and the number of decoding iterations as shown in Fig. 1(b) (denoted as I). The parameter E directly impacts the hardware resource consumption of the decoder. As demonstrated in [33], [34], the amount of RAM required to store extrinsic information and the number of interconnects between CNUs and VNUs increase with E . Additionally, in [35], LDPC decoders with different $\mathbf{H}$ density (defined as E divided by the total number of elements in $\mathbf{H}$) were implemented on a Field Programmable Gate Array (FPGA). The results indicate that the utilization of FPGA resources, including LUTs, FFs, and RAMs, shows a roughly linear relationship with E . Similarly, [21] suggests that the decoder complexity increases approximately linearly with the number of non-zero entries. Based on the analyses and experimental results reported in [21], [33], [34], [35], a rough estimate of the decoder's hardware resource usage is adopted as

$$W = \alpha E \quad (1)$$

where α is a constant related to the specific decoder implementation. Depending on implementation focus, W can be measured in terms of LUTs, FFs, RAMs, or their combination. On the other hand, for a single-rate LDPC code with an information block length of K , the throughput of LDPC decoders can be

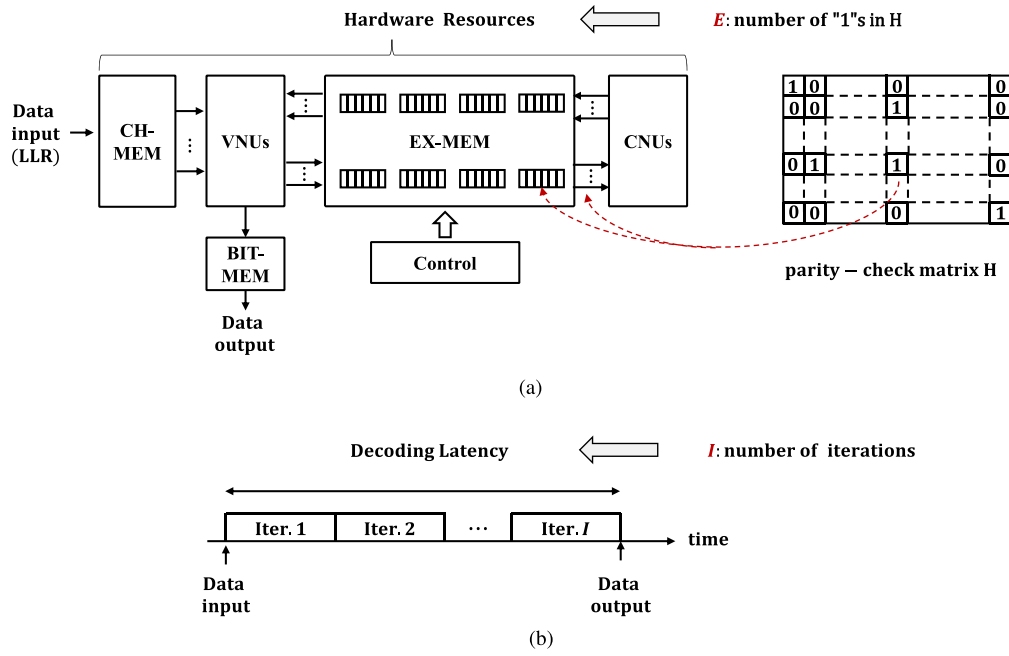


Fig. 1. Illustration of the two key factors in throughput-optimized LDPC implementation. (a) Hardware resources usage affected by the number of non-zero elements in the parity-check matrix. (b) Decoding latency affected by the number of decoding iterations.

expressed as [36, Ch. 6]

$$T = \frac{Kf}{IS} \quad (2)$$

where f is the decoder clock frequency, I is the number of decoding iterations, and S is the number of clock cycles per iteration. In certain high-speed communication scenarios, multi-core parallel decoders [37], [38] are often employed to achieve higher decoding throughput. Minimizing the resource usage of a single decoder allows for a greater number of parallel cores in a resource-limited system, and this can significantly enhance the normalized decoding throughput [35]. Therefore, it is essential to consider not only the throughput T of a single-core LDPC decoder but also its hardware resource usage W . Taking into account both hardware resource usage and decoding throughput, we define the normalized throughput as a performance metric that reflects the achievable throughput per unit of resource usage, which is formally given by

$$\gamma = \frac{T}{W} = \frac{1}{\alpha E} \times \frac{Kf}{IS} \quad (3)$$

The unit of the normalized throughput γ is expressed as bps per unit resource.¹ In (3), α and f are primarily related to the specific decoder architecture, while S is mainly determined by the used decoding algorithm. If the same decoding algorithm and decoder architecture are adopted, α , f and S remain approximately constant for LDPC codes with different parity-check matrices. This leads to a simplified measure of normalized

throughput for single-rate LDPC as

$$\tilde{\gamma} = \frac{1}{E} \times \frac{K}{I} \quad (4)$$

As shown in (4), the desired throughput-optimized LDPC codes should exhibit two key characteristics. First of all, they should demonstrate excellent decoding performance, requiring fewer iterations I to achieve a given error rate. Secondly, their parity-check matrices should be sufficiently sparse, characterized by a smaller E value, to reduce hardware resource consumption. While conventional threshold-optimized codes typically focus only on decoding performance, throughput-optimized codes introduce matrix sparsity as an additional optimization parameter, thereby increasing the complexity of the design and optimization process. The authors of [20] adopted a similar design strategy for single-rate irregular LDPC codes. The codes they designed show a significant reduction in decoding complexity compared to throughput-optimized codes, resulting in higher data throughput, while still maintaining excellent decoding performance. In the subsequent sections, we demonstrate that this design strategy is also effective for multi-rate LDPC codes.

B. Proposed Multi-Rate Weighted Normalized Throughput Metric for LDPC Codes

In satellite communication systems, channel coding schemes must support flexible code rates due to varying channel conditions and data rate requirements for different terminals. Therefore, this paper focuses on the design of multi-rate throughput-optimized LDPC codes, which is more challenging than single-rate code construction. In particular, multi-rate codes with a nested parity-check matrix structure are adopted, which allows multiple code rates to share a single decoder.

¹Depending on the design needs, resource units may include LUTs, FFs, RAMs, etc.

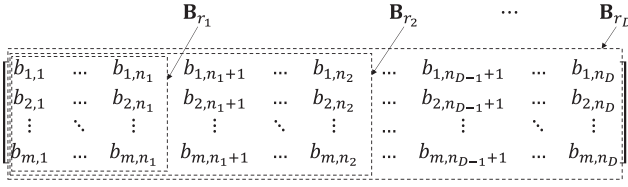


Fig. 2. Structure of nested multi-rate P-LDPC codes protomatrix obtained through lengthening.

We begin by presenting the multi-rate weighted normalized throughput metric. Suppose the code ensemble $\mathcal{C}$ to be designed has D code rates $r_1, \dots, r_D$. The corresponding parity-check matrices are $\mathbf{H}_{r_1}, \dots, \mathbf{H}_{r_D}$ in a nested structure, with the largest matrix denoted as $\mathbf{H}$. We follow the same strategy as in (4) to define the normalized throughput for multi-rate LDPC codes, which is the ratio of decoding throughput to hardware resource usage. However, the case is different for multi-rate codes, which requires a more detailed discussion.

Firstly, for the resource factor, the parameter E in (4) is replaced by E_{total} , where E_{total} represents the total number of non-zero elements in the entire parity-check matrix $\mathbf{H}$. This change is due to the fact that all code rates share the same decoder in practical hardware implementations of LDPC codes with a nested structure, and the resource consumption of the decoder is proportional to E_{total} . Secondly, for the throughput factor, the K/I term is replaced by a weighted sum of the decoding throughput at each code rate, represented as $\sum_{d=1}^D w_d \frac{K_d}{I_d}$, where w_d is the weighting factor, which can be assigned as the usage frequency of rate r_d in practice. Here, K_d and I_d denote the information block length and the number of decoding iterations for rate r_d , respectively. Consequently, as an extension of the single-rate normalized throughput metric, the weighted normalized throughput metric for multi-rate LDPC codes is defined as follows.

$$\tilde{\gamma}(\mathcal{C}) = \frac{1}{E_{\text{total}}} \times \sum_{d=1}^D w_d \frac{K_d}{I_d} \quad (5)$$

The key idea of the proposed metric is to link the hardware resource consumption of the entire code ensemble through E_{total} . Using the proposed metric as the optimization objective during the code construction ensures a lower overall hardware implementation complexity. Additionally, the decoding iterations for each code rate are incorporated into the metric through a weighted sum, allowing for the fulfillment of different code design requirements.

This paper primarily examines the multi-rate protograph LDPC (P-LDPC) codes obtained through lengthening with a protomatrix structure as depicted in Fig. 2. These codes are widely used in satellite communication scenarios, such as the accumulate-repeat-4-jagged-accumulate (AR4JA) codes [23] in the CCSDS standard. The rate- r_d code has a protomatrix $\mathbf{B}_{r_d}$ of size $m \times n_d$ ($d \in \{1, 2, \dots, D\}$), with elements that are non-negative integers, potentially greater than one. The latter protomatrix $\mathbf{B}_{r_{d+1}}$ can be constructed by adding $n_{d+1} - n_d$ columns to $\mathbf{B}_{r_d}$, keeping the same number of rows. The code rate

r_d is computed as $r_d = 1 - m/n_d$. Let the largest protomatrix be denoted as $\mathbf{B}$, where $\mathbf{B} = \mathbf{B}_{r_D} = \{b_{i,j}\}_{i=1,\dots,m,j=1,\dots,n_D}$. Using the two-step lifting method [39] to construct the entire parity-check matrix $\mathbf{H}$, $\mathbf{B}$ is first lifted by a small factor L_1 to eliminate the multi-edges and obtain a base matrix containing only 1 or 0 entries. This base matrix is then lifted by a factor L_2 , replacing the 1's with circulant permutation matrix (CPM) and 0's with zero matrices to obtain the final parity-check matrix $\mathbf{H}$. Consequently, it follows that $E_{\text{total}} = L_1 L_2 \times \sum_{i,j} b_{i,j}$ and $K_d = L_1 L_2 \times k_d$, where $k_d = n_d - m$. The metric in (5) for two-step-lifting multi-rate P-LDPC codes can be further expressed as follows.

$$\tilde{\gamma}(\mathcal{C}_{\mathbf{B}}) = \frac{1}{\sum_{i,j} b_{i,j}} \times \sum_{d=1}^D w_d \frac{k_d}{I_d} \quad (6)$$

The proposed multi-rate weighted normalized throughput metric and optimization techniques in this paper are also applicable to other types of multi-rate P-LDPC codes, such as rate-compatible (RC) codes that feature a raptor-like protomatrix structure [40], [41]. Since we primarily focus on multi-rate LDPC codes constructed through lengthening, the optimization results for RC LDPC codes are not discussed in this paper.

III. DESIGN OF THROUGHPUT-OPTIMIZED MULTI-RATE LDPC CODES

In this section, we first introduce the protograph extrinsic information transfer (PEXIT) algorithm, which is used to estimate the required number of decoding iterations. Next, we formulate the optimization problem for constructing throughput-optimized multi-rate LDPC codes, using the proposed weighted normalized throughput metric as the optimization objective. Finally, a protomatrix search method based on the differential evolution (DE) algorithm is proposed to find the codes that maximize the multi-rate weighted normalized throughput.

A. PEXIT Analysis for Iteration Estimation

The PEXIT algorithm [42] is widely used for analyzing the threshold performance of protograph-based LDPC due to its low computational complexity and accurate threshold predictions. In this work, the PEXIT algorithm is employed to estimate the number of decoding iterations I_d under a constrained decoding threshold. Specifically, for a given protomatrix $\mathbf{B}$, the function $f_{\text{PEXIT}}(\mathbf{B}, \eta^*)$ returns the required PEXIT iterations to achieve the target energy per bit to noise power spectral density ratio (E_b/N_0) value η^* . The parameter I_d for code rate r_d is then determined as follows.

$$I_d = f_{\text{PEXIT}}(\mathbf{B}_{r_d}, \eta_d^*), d = 1, \dots, D \quad (7)$$

where η_d^* is the constrained decoding threshold for rate r_d .

For a protomatrix $\mathbf{B} = \{b_{i,j}\}_{i=1,\dots,m,j=1,\dots,n}$ with code rate $R = m/n$, let C_i denote the i -th check node and V_j denote the j -th variable node, corresponding to the i -th row and the j -th column of $\mathbf{B}$, respectively. The steps of $f_{\text{PEXIT}}(\mathbf{B}, \eta^*)$ for computing the required iterations I under a constrained

decoding threshold η^* using the PEXIT algorithm are described as follows [42]:

1) Initialization:

For $j = 1, \dots, n$, the channel mutual information (MI) is computed as:

$$I_{ch,j} = 1 - \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi\sigma_0^2}} e^{-\frac{(l-\sigma_0^2/2)^2}{2\sigma_0^2}} \cdot \log_2(1 + e^{-l}) dl$$

$$= J(\sigma_0) \quad (8)$$

where the parameter σ_0 is defined as:

$$\sigma_0 = 8R\eta^* \quad (9)$$

For $i = 1, \dots, m$ and $j = 1, \dots, n$, the *a priori* MI from C_i to V_j is initialized as $I_{A,V}^{i \rightarrow j} = 0$.

2) Variable nodes to check nodes update:

For $i = 1, \dots, m$ and $j = 1, \dots, n$, the *extrinsic* MI from V_j to C_i is computed as:

$$I_{E,V}^{j \rightarrow i} = J \left(\sqrt{(b_{i,j} - 1)[J^{-1}(I_{A,V}^{i \rightarrow j})]^2 + [J^{-1}(I_{ch,j})]^2} \right. \\ \left. + \sum_{i' \neq i} b_{i',j} [J^{-1}(I_{A,V}^{i' \rightarrow j})]^2 \right) \quad (10)$$

and the *a priori* MI from V_j to C_i is updated as $I_{A,C}^{j \rightarrow i} = I_{E,V}^{j \rightarrow i}$.

3) Check nodes to variable nodes update:

For $i = 1, \dots, m$ and $j = 1, \dots, n$, the *extrinsic* MI from C_i to V_j is computed as:

$$I_{E,C}^{i \rightarrow j} = 1 - J \left(\sqrt{(b_{i,j} - 1)[J^{-1}(1 - I_{A,C}^{j \rightarrow i})]^2} \right. \\ \left. + \sum_{j' \neq j} b_{i,j'} [J^{-1}(1 - I_{A,C}^{j' \rightarrow i})]^2 \right) \quad (11)$$

and the *a priori* MI from C_i to V_j is updated as $I_{A,V}^{i \rightarrow j} = I_{E,C}^{i \rightarrow j}$.

4) APP-LLR MI update:

For $j = 1, \dots, n$, the MI between the log-likelihood ratio and the codeword bit is computed as:

$$I_{APP}^j = J \left(\sqrt{\sum_i b_{i,j} [J^{-1}(I_{A,V}^{i \rightarrow j})]^2 + [J^{-1}(I_{ch,j})]^2} \right) \quad (12)$$

5) Judgement:

- If the number of iterations does not exceed the maximum allowed iterations and $I_{APP}^j = 1, \forall j$, decoding under the constrained threshold η^* is successfully completed, and the current PEXIT iterations number is returned. Otherwise, the algorithm proceeds to step 2 for further iterations.
- If the iteration count exceeds the maximum iterations and $I_{APP}^j \neq 1, \exists j$, decoding fails under the constrained threshold η^* , and a predefined large integer is returned.

In conventional threshold-optimized LDPC design, where the number of PEXIT iterations is predetermined, a comprehensive search is performed to find the minimum value of η^* that allows the PEXIT algorithm to successfully converge, thereby determining the accurate decoding threshold. This process involves repeated evaluations of the function $f_{\text{PEXIT}}(\mathbf{B}, \eta^*)$, resulting in relatively high complexity. In contrast, this work uses the function $f_{\text{PEXIT}}(\mathbf{B}, \eta^*)$ only once to calculate the required PEXIT iterations for a constrained decoding threshold, thereby significantly reducing the computational complexity.

B. Problem Formulation

By maximizing the multi-rate weighted normalized throughput metric defined in (6), throughput-optimized multi-rate P-LDPC codes with low hardware resource utilization and fast decoding convergence can be obtained. To ensure the decoding performance under low iterations, a relatively small upper limit, $I_{\max}$, is imposed on I_d , i.e., $I_d \leq I_{\max}, d = 1, \dots, D$. Furthermore, as reported in [23], to guarantee the linear minimum distance growth property of designed codes, the weights of the columns to be optimized in $\mathbf{B}$ are constrained to be at least $d_{v,\min} = 3$. Let these columns be denoted as vectors $\mathbf{b}_u$ for $u = 1, \dots, U$. The column weight constraints can then be expressed by inequalities $\mathbf{1}^T \mathbf{b}_u - d_{v,\min} \geq 0, u = 1, \dots, U$.

In summary, suppose the columns to be optimized in $\mathbf{B}$ are $\mathbf{b}_u, u = 1, \dots, U$, the optimization problem is defined with the following inputs: the constrained decoding thresholds η_d^* and weighting factors w_d (for $d = 1, \dots, D$), the upper limit $I_{\max}$ on the number of iterations. The problem is formulated as follows.

$$\max_{\mathbf{B}} \quad \tilde{\gamma}(\mathbf{C}_B) = \frac{1}{\sum_{i,j} b_{i,j}} \times \sum_{d=1}^D w_d \frac{k_d}{I_d} \quad (13)$$

$$\text{s.t.} \quad I_d \leq I_{\max}, \quad d = 1, \dots, D \\ \mathbf{1}^T \mathbf{b}_u - d_{v,\min} \geq 0, \quad u = 1, \dots, U \quad (14)$$

where the required decoding iterations I_d under the constrained threshold η_d^* is calculated using the PEXIT algorithm according to (7).

Unlike previous studies, such as those by Tang et al. [21], which primarily concentrated on optimizing decoding thresholds while manually restricting E and I , the proposed optimization problem uses the multi-rate weighted normalized throughput as the objective. This allows the optimization to simultaneously target both high throughput and low resource consumption. Meanwhile, the decoding thresholds are treated as performance constraints in (7). Experimental results in Section IV and Section V show that this design strategy can effectively obtain throughput-optimized codes with minimal resource usage, while maintaining comparable decoding performance to conventional threshold-optimized methods.

C. Protomatrix Search Based on DE Algorithm

The optimization problem presented in Section IV-A is non-convex and involves multiple discrete variables. The size of its search space grows exponentially with the number of variables, making a brute-force search infeasible. To address this, we

employ the differential evolution (DE) algorithm [29], a genetic algorithm that produces approximate optimal solutions through successive evolutionary steps. The DE algorithm is particularly effective for handling complex problems with many parameters, such as protomatrix optimization. During each evolutionary iteration, protomatrices with higher multi-rate weighted normalized throughput scores are selected, gradually leading to the development of throughput-optimized codes over multiple generations. The main steps of the proposed algorithm are outlined as follows.

GIVEN: the number of code rates D ; the Q integer variables $x_q (q = 1, 2, \dots, Q)$ in the $m \times n_D$ protomatrix $\mathbf{B}$; the non-negative minimum and maximum values of the variables $x_{\min}$ and $x_{\max}$; the minimum column weight $d_{v,\min}$; the constrained decoding thresholds vector $\boldsymbol{\eta}^* = (\eta_1, \dots, \eta_D)$; the weighting factors w_d for $d = 1, \dots, D$; the upper limit of PEXIT iterations $I_{\max}$; the number of candidate matrices S and the number of generations G ; the crossover probability p_c .

1) Initialization:

For $s = 1, \dots, S$, generate the initial candidate protomatrix $\mathbf{B}_s^{(0)}$ with the Q integer variables randomly chosen from $[x_{\min}, x_{\max}]$.

2) Mutation:

For $s = 1, \dots, S$, $q = 1, \dots, Q$, the q -th variable of the s -th mutation matrix $\mathbf{M}_s^{(g)}$ in generation g is defined by

$$\mathbf{M}_s^{(g)}(q) = \Phi(\mathbf{B}_{t_1}^{(g)}(q) + 0.5(\mathbf{B}_{t_2}^{(g)}(q) - \mathbf{B}_{t_3}^{(g)}(q)))$$

where t_1, t_2, t_3 are three different integers randomly chosen from $[1, S]$. The function $\Phi(x)$ replaces x with the closest integer in $[x_{\min}, x_{\max}]$.

3) Crossover:

For $s = 1, \dots, S$, $q = 1, \dots, Q$, the q -th variable of the s -th new candidate matrix $\mathbf{N}_s^{(g)}$ is generated by

$$\mathbf{N}_s^{(g)}(q) = \begin{cases} \mathbf{M}_s^{(g)}(q) & \text{, with probability } p_c, \\ \mathbf{B}_s^{(g)}(q) & \text{, otherwise.} \end{cases}$$

4) Selection:

For $s = 1, \dots, S$, candidate protomatrices for the $(g+1)$ -th generation are selected according to the rule:

$$\mathbf{B}_s^{(g+1)} = \begin{cases} \mathbf{N}_s^{(g)} & \text{, if } \tilde{\gamma}(\mathcal{C}_{\mathbf{N}_s^{(g)}}) \mathbb{I}(\mathbf{N}_s^{(g)}, \mathbf{B}_s^{(g)}) \geq \tilde{\gamma}(\mathcal{C}_{\mathbf{B}_s^{(g)}}), \\ \mathbf{B}_s^{(g)} & \text{, otherwise.} \end{cases}$$

where the indicator function $\mathbb{I}(\mathbf{N}_s^{(g)}, \mathbf{B}_s^{(g)})$ returns 1 if the number of satisfied inequality constraints in (14) of $\mathbf{N}_s^{(g)}$ is not less than $\mathbf{B}_s^{(g)}$ and otherwise returns 0.

5) Termination:

Step 2-4 are executed for G generations and the candidate protomatrix in the G -th generation that satisfies all the constraints and gives the highest multi-rate weighted normalized throughput is selected as the final solution for $\mathbf{B}$.

Instead of lengthening the protomatrix rate-by-rate [24], [43], the proposed search algorithm optimizes the entire protomatrix directly to maximize the multi-rate weighted normalized throughput, leading to significant reductions in both decoding iterations and hardware resource consumption.

TABLE II
CODE RATE EMPHASIS AND WEIGHTING FACTORS FOR DIFFERENT SCENES

Scene	Code Rate Emphasis	Weighting Factors (w_1, w_2, w_3, w_4)
A	Equal importance	(0.25, 0.25, 0.25, 0.25)
B	Low-rate dominant	(0.5, 0.3, 0.15, 0.05)
C	High-rate dominant	(0.05, 0.15, 0.3, 0.5)

IV. CODE DESIGN EXAMPLE AND PERFORMANCE EVALUATION BASED ON PEXIT ANALYSIS

In this section, we present designing examples of specific four-rate throughput-optimized P-LDPC codes using the proposed code optimization method. Theoretical analyses and comparisons of the decoding performance and throughput based on the PEXIT algorithm are also provided.

A. Protomatrix Design

As a demonstration, we design multi-rate P-LDPC codes with code rates 1/2, 2/3, 3/4 and 4/5, the same rate setting as the AR4JA code in the CCSDS standard [44], following the code structure illustrated in Fig. 2. Significantly, the non-punctured code structure [43] is adopted which offers faster decoding convergence than punctured codes. The optimization target $\mathbf{B}$ has a size of 4×20 . As suggested in [45], three columns of the protomatrix are fixed as weight-2 columns for better performance. To limit the search space, the maximum value of the entries is set to 3. Thus, $\mathbf{B}$ has the form as

$$\mathbf{B} = \begin{bmatrix} 1 & 0 & 0 & x_1 & x_2 & \dots & x_{16} & x_{17} \\ 0 & 1 & 0 & x_{18} & x_{19} & \dots & x_{33} & x_{34} \\ 0 & 0 & 1 & x_{35} & x_{36} & \dots & x_{50} & x_{51} \\ 1 & 1 & 1 & x_{52} & x_{53} & \dots & x_{67} & x_{68} \end{bmatrix} \quad (15)$$

In (15), the optimization target $\mathbf{B}$ has 68 integer variables x_q , where $q = 1, \dots, 68$ and $0 \leq x_q \leq 3$. The minimum column weight constraints are expressed as $\sum_{l=0}^3 x_{u+17l} \geq 3$ for $u = 1, 2, \dots, 17$ with $d_{v,\min} = 3$. Optimization is carried out with $I_{\max} = 25$, $\boldsymbol{\eta}^* = (1.25, 1.83, 2.27, 2.61)$ dB, $p_c = 0.3$, $S = 700$ and $G = 200$. The constrained threshold vector $\boldsymbol{\eta}^*$ is selected based on the decoding thresholds of Nguyen's delay-limited code [24] under 20 decoding iterations. As a threshold-optimized code, it exhibits low decoding thresholds across all rates. This constrained threshold setting serves as a well-performing baseline, while the relaxed iteration constraint $I_{\max} = 25$ leaves room to improve decoding throughput and reduce resource usage during optimization.

With regard to the weighting factors w_d , three scenes (Scene A, B and C) with different weighting values are considered in code design and performance evaluation to demonstrate the applicability of our proposed method, as summarized in Table II. Scene A assigns equal importance to all code rates, representing a balanced usage case. Scene B reflects a low-rate dominant setting, corresponding to small user terminals with limited power and poor channel conditions. In contrast, Scene C emphasizes high-rate usage, suited for scenarios with higher transmit power and better channel conditions. The optimized multi-rate codes for Scene A, B and C are denoted as $\mathcal{C}_A$, $\mathcal{C}_B$ and $\mathcal{C}_C$, respectively.

TABLE III
MULTI-RATE WEIGHTED NORMALIZED THROUGHPUT $\tilde{\gamma}$ AND RESOURCE ESTIMATE E_{TOTAL} COMPARISON WHEN ACHIEVING DIFFERENT PEXIT DECODING THRESHOLDS¹ FOR CODES WITH THE SAME LENGTH UNDER DIFFERENT SCENES²

Codes	Resource Estimate ³ E_{total}	Required PEXIT Iterations (I_1, I_2, I_3, I_4)	Multi-Rate Weighted Normalized Throughput $\tilde{\gamma}(\times 10^{-3})$		
			Scene A	Scene B	Scene C
AR4JA Code [23]	$78L_1L_2$	(26, 28, 29, 30)	4.44	3.22	5.66
Nguyen's Code [24]	$93L_1L_2$	(20, 20, 20, 20)	5.38	3.76	6.99
Proposed C_A	$75L_1L_2$	(21, 21, 22, 22)	6.15	4.37	7.92
Proposed C_B	$72L_1L_2$	(20, 20, 23, 27)	5.95	4.55	7.26
Proposed C_C	$73L_1L_2$	(24, 26, 21, 22)	6.07	4.08	8.07

¹ The four-rates codes are evaluated to achieve thresholds $\eta^* = (1.25, 1.83, 2.27, 2.61)$ dB.

² Weighting factors (w_1, w_2, w_3, w_4) for Scene A, Scene B and Scene C are (0.25, 0.25, 0.25, 0.25), (0.5, 0.3, 0.15, 0.05) and (0.05, 0.15, 0.3, 0.5), respectively.

³ L_1 and L_2 are the two-step lifting factors in parity-check matrix construction.

The corresponding protomatrices $\mathbf{B}_{C_A}$, $\mathbf{B}_{C_B}$ and $\mathbf{B}_{C_C}$ after an extensive search using the DE algorithm are given in (16), (17) and (18) shown at the bottom of this page. The protomatrix for code rate 1/2, 2/3, 3/4, 4/5 can be obtained by selecting the first 8, 12, 16, and 20 columns of the entire protomatrix, respectively.

We choose the AR4JA code [23] and Nguyen's delay-limited code [24] as benchmark codes for comparison, since the former is a widely adopted standard design and the latter is specifically optimized for minimizing decoding iterations. For C_A , C_B , C_C , the AR4JA code and Nguyen's code, the hardware resource consumption estimates E_{total} are $75L_1L_2$, $72L_1L_2$, $73L_1L_2$, $78L_1L_2$, $93L_1L_2$, respectively.² This implies that all three of our designed codes exhibit lower decoder hardware usage than the benchmark codes. Specifically, E_{total} is reduced by 19.4%–22.6% compared to Nguyen's code and 3.8%–7.7% compared to the AR4JA code, with C_B achieves the lowest usage. However, to achieve the same decoding thresholds, our codes generally require slightly more iterations than Nguyen's code in exchange for improved weighted normalized throughput. The decoding threshold versus iteration performance and the normalized throughput comparison are presented in Fig. 3 and Table III, respectively. This trade-off is intentional and aligns with our

primary objective of maximizing normalized throughput under constrained hardware resources.

B. Theoretical Performance Analysis Based on PEXIT

This section presents further theoretical analyses of the three optimized codes based on PEXIT, including an evaluation of multi-rate weighted normalized throughput under different threshold performance requirements and a comparison of decoding thresholds under varying iterations.

A comparison of the multi-rate weighted normalized throughput $\tilde{\gamma}$ of the five codes is shown in Table III. The constrained threshold vector η^* in Section IV-A is selected as the required threshold values. The highest $\tilde{\gamma}$ value for each scene is marked in bold. As shown in Table III, the proposed three codes consistently outperform the two benchmark codes in terms of multi-rate weighted normalized throughput $\tilde{\gamma}$ in all scenes. While the total resource estimates E_{total} for C_A , C_B , and C_C remain at a similar and relatively low level, their required PEXIT iterations (I_1, I_2, I_3, I_4) vary noticeably, resulting in different throughput outcomes in the three scenes. Among them, C_A maintains near-optimal iteration counts across all code rates, which contributes to its generally favorable throughput performance in all scenes. Compared to Nguyen's code, C_A achieves $\tilde{\gamma}$ gains of 14.31% in Scene A, 16.22% in Scene B, and 13.30% in Scene C. In comparison with the AR4JA code, the corresponding improvements rise to 38.31%, 35.71%, and 39.93%, respectively. In contrast, C_B , which is particularly optimized for

$$\mathbf{B}_{C_A} = \begin{bmatrix} 3 & 2 & 0 & 1 & 0 & 1 & 0 & 0 \\ 3 & 3 & 0 & 2 & 0 & 0 & 1 & 0 \\ 3 & 3 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 2 & 0 & 3 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 & 0 \\ 0 & 2 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 0 & 1 & 2 & 1 \\ 1 & 0 & 3 & 1 \\ 0 & 2 & 1 & 1 \\ 2 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 2 & 2 \end{bmatrix} \quad (16)$$

$$\mathbf{B}_{C_B} = \begin{bmatrix} 3 & 1 & 0 & 2 & 0 & 1 & 0 & 0 \\ 3 & 2 & 1 & 1 & 0 & 0 & 1 & 0 \\ 3 & 1 & 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 2 & 1 & 2 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 & 1 & 0 \\ 2 & 1 & 1 & 0 \\ 3 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 2 & 2 & 2 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 2 \\ 0 & 2 & 0 & 0 \\ 2 & 0 & 1 & 0 \\ 1 & 1 & 2 & 1 \end{bmatrix} \quad (17)$$

$$\mathbf{B}_{C_C} = \begin{bmatrix} 3 & 0 & 2 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 3 & 0 & 0 & 1 & 0 \\ 1 & 0 & 2 & 1 & 1 & 0 & 0 & 1 \\ 0 & 2 & 0 & 0 & 2 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 & 0 & 1 \\ 2 & 0 & 1 & 0 \\ 1 & 0 & 2 & 0 \\ 0 & 3 & 0 & 2 \end{bmatrix} \begin{bmatrix} 3 & 2 & 0 & 1 \\ 2 & 3 & 2 & 0 \\ 2 & 3 & 1 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 2 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 2 \end{bmatrix} \quad (18)$$

²The hardware resource consumption estimate E_{total} is compared with the same transmitted code length for the three codes. Nguyen's code and the proposed codes are lifted by L_1L_2 , while the AR4JA code is lifted by $2L_1L_2$ to match the code lengths of the other two codes, as it uses a 3×11 protomatrix with one column for puncturing.

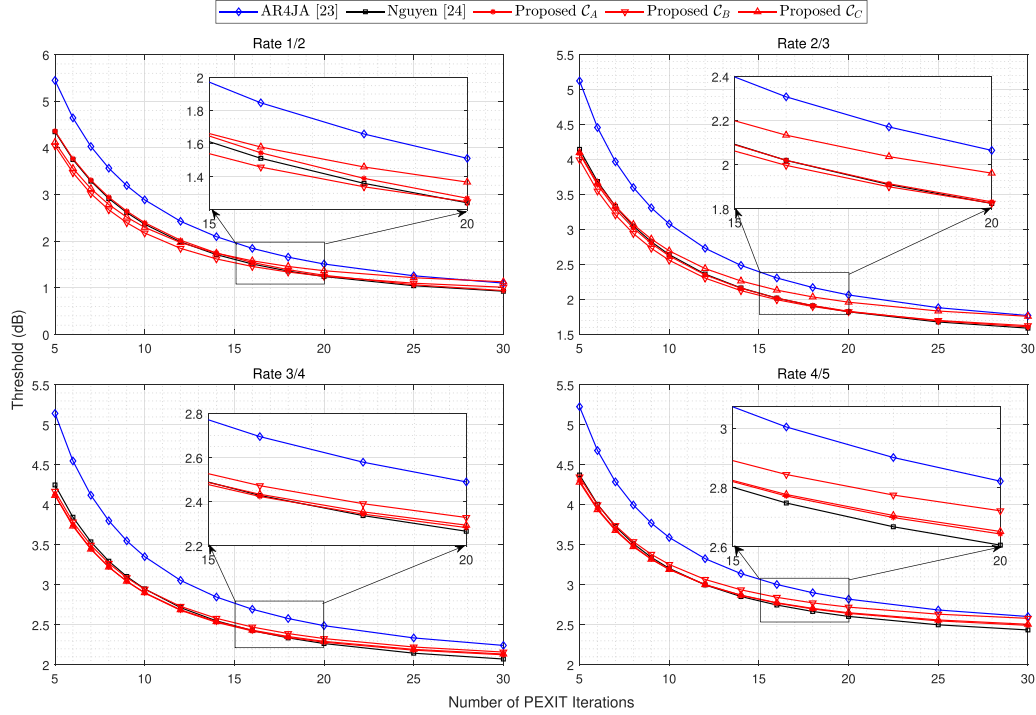


Fig. 3. Decoding thresholds comparison under different number of PEXIT iterations.

low-rate-dominated scenes, requires the fewest iterations at code rate 1/2 and 2/3. As a result, it achieves the highest normalized throughput in Scene B with $\tilde{\gamma}$ improvements of 21.01% over Nguyen's code and 41.30% over the AR4JA code. Finally, C_C achieves the highest normalized throughput in Scene C. It yields a 15.45% improvement over Nguyen's code and a 42.58% gain compared to the AR4JA code. These results confirm that the proposed codes, each tailored to a specific scene, effectively achieve the highest multi-rate weighted normalized throughput in their respective target scenes.

In Fig. 3, a comparison of decoding thresholds is presented, with detailed views of iterations 15–20 in each subplot. Among all considered codes, Nguyen's code still achieves the optimal decoding thresholds in most cases, whereas our codes require slightly more iterations to reach similar thresholds as a result of maximizing normalized throughput. C_A maintains performance close to Nguyen's across all rates, making it a strong general-purpose choice. C_B yields the lowest decoding thresholds among the three proposed codes under low code rates, even surpassing Nguyen's code at rate-1/2. C_C , on the other hand, performs best among the proposed codes at high code rates, approaching the optimal results. The observed performance trends confirm the effectiveness of the tailored design for different code rate usage distributions.

V. NUMERICAL SIMULATION AND PERFORMANCE COMPARISON

In this section, practical LDPC codes are constructed for numerical evaluation over different channel models. The proposed protomatrix, given in (16)–(18), is lifted with $L_1 = 4$

to remove parallel edges using the progressive edge growth (PEG) algorithm [46]. The resulting base matrix is then lifted with $L_2 = 256$ using the simulated annealing method [47] to construct parity check matrices. For a fair comparison, both Nguyen's delay-limited code and the AR4JA code are lifted using the same algorithms. Nguyen's code is lifted with the same factors, while the AR4JA code is lifted with $L'_1 = L_1 = 4$ and $L'_2 = 2L_2 = 512$ to ensure equal information lengths and transmitted block lengths. The information lengths of rates 1/2, 2/3, 3/4, and 4/5 are 4096, 8192, 12288, and 16384, respectively.

A. Channel Assumptions and Parameters

Three types of satellite channels are considered in this article, i.e., the additive white Gaussian noise (AWGN) channel, the Rician fading channel and the Doppler-affected channel with frequency offsets, with BPSK as the modulation scheme.

In particular, as discussed in detail in [48], [49], the partial Rician fading channel model for coded-interleaving systems is adopted. A segment length of $s = 100$ bits is used in the simulation, with each segment experiencing independent fading. The Rician K factor (defined as the power ratio between the line-of-sight component and the scattered components) is set to 5 dB to represent moderate fading conditions.

To evaluate the impact of Doppler effects in high-mobility satellite scenarios, we consider a normalized frequency offset (NFO) of $\Delta f_d T_s = 1 \times 10^{-4}$, following the typical setting in [50], [51]. In simulation, the transmitted symbol $x(n)$ is rotated by a phase term $e^{-j2\pi n \Delta f_d T_s}$ over a frame of 4096 symbols. The NFO is estimated and compensated before LDPC decoding using the widely adopted expectation-maximization (EM) algorithm tailored for iterative decoders [50].

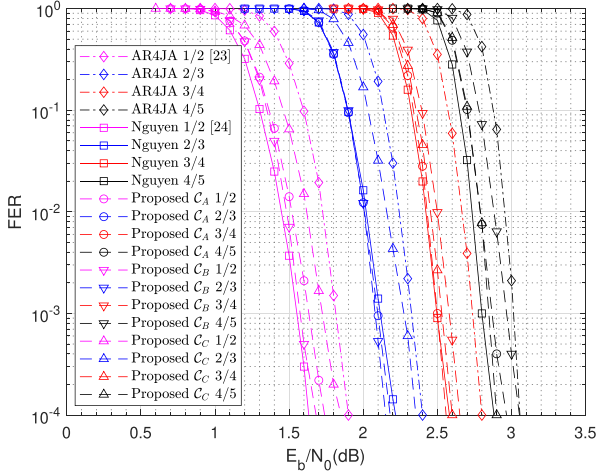


Fig. 4. FER performance comparison over an AWGN channel with fixed iterations = 20.

B. FER Performance Analysis

The frame error rate (FER) performance of the two benchmark codes and the three proposed codes is firstly evaluated. To ensure fairness, the number of decoding iterations is fixed as 20 in the simulation, as Nguyen's delay-limited code is specifically optimized for this iteration count.

Fig. 4 shows the FER performance of the proposed codes and the two benchmark codes under belief-propagation (BP) [36] decoding. C_A shows an improvement of approximately 0.2 dB over the AR4JA code across all four code rates at a target FER of 10^{-3} . Compared to Nguyen's code, however, a performance gap of about 0.1 dB is observed at code rate 1/2 and 4/5. On the other hand, C_B demonstrates near-optimal performance at low code rates. However, its performance degrades at higher rates, with gaps of 0.10 dB and 0.16 dB observed at rates 3/4 and 4/5, respectively. In contrast, C_C exhibits a complementary trend, achieving FER performance within 0.05 dB of Nguyen's code at high rates (3/4 and 4/5), but showing larger gaps at rates 1/2 and 2/3.

The FER performance over partial Rician channels with $K = 5$ dB is presented in Fig. 5. Compared to the AWGN case, the FER curves are noticeably flatter, which is consistent with the results reported for partial Rician fading channels in [49]. At a target FER of 10^{-3} , the required E_b/N_0 increases by approximately 1.9–2.8 dB across all code rates relative to the AWGN results. In addition, FER results under Doppler-affected channels with NFO = 1×10^{-4} are shown in Fig. 6, with performance degradations of approximately 0.3–0.4 dB for the four code rates compared to the AWGN case.

Although the absolute FER performance of each code varies under different channel conditions, the relative advantages of the proposed three codes remain consistent. Specifically, C_B remains highly competitive at rate 1/2 and 2/3, with its performance closely approaching that of Nguyen's code. In contrast, C_C demonstrates near-optimal performance at rate 3/4 and 4/5, with its performance consistently within 0.1 dB of the optimal result under each channel condition. As expected, C_A achieves

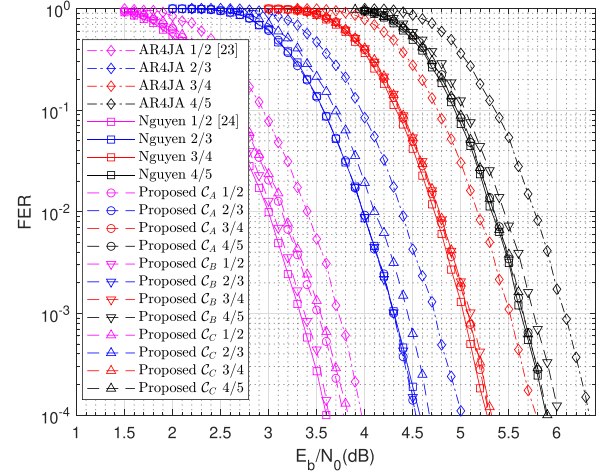


Fig. 5. FER performance comparison over a partial Rician fading channel ($K = 5$ dB) with fixed iterations = 20.

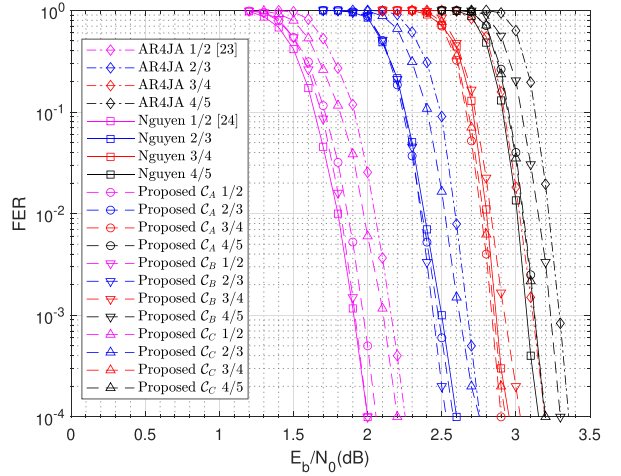


Fig. 6. FER performance comparison over a Doppler-affected channel with a normalized frequency offset $\Delta f_d T_s = 1 \times 10^{-4}$ with fixed iterations = 20.

the most balanced performance across all rates, with its FER results remaining close to the optimal and generally falling between those of C_B at low code rates and C_C at high code rates. The results demonstrate that even under different types of non-Gaussian fading, the proposed codes continue to exhibit their intended performance advantages.

C. Multi-Rate Weighted Normalized Throughput Performance Comparison

In this section, the multi-rate weighted normalized throughput $\tilde{\gamma}$ is thoroughly examined and compared under different channel conditions based on simulation results. To distinguish it from the theoretical estimate of I_d derived from the PEXIT analysis in Section IV, for code rate r_d , we denote by $I_d^{\text{sim}}(p_e, \xi)$ (abbreviated as I_d^{sim}) the required number of BP decoding iterations to achieve a target FER of p_e at a bit energy-to-noise ratio (E_b/N_0) of ξ in simulations following the notation in [52]. In practice, I_d^{sim} is obtained by gradually reducing the maximum decoding

TABLE IV
REQUIRED DECODING ITERATIONS ($I_1^{\text{sim}}, I_2^{\text{sim}}, I_3^{\text{sim}}, I_4^{\text{sim}}$) COMPARISON¹

Codes	Required Decoding Iterations ($I_1^{\text{sim}}, I_2^{\text{sim}}, I_3^{\text{sim}}, I_4^{\text{sim}}$)		
	AWGN	Partial Rician $K = 5$ dB	Doppler-affected $\Delta f_d T_s = 1 \times 10^{-4}$
AR4JA Code [23]	(28, 27, 31, 29)	(28, 33, 34, 33)	(27, 26, 29, 25)
Nguyen's Code [24]	(20, 20, 20, 20)	(20, 20, 20, 20)	(20, 20, 20, 20)
Proposed $\mathcal{C}_A$	(23, 20, 21, 24)	(24, 20, 21, 22)	(23, 19, 20, 22)
Proposed $\mathcal{C}_B$	(22, 19, 23, 28)	(21, 20, 23, 27)	(20, 19, 23, 26)
Proposed $\mathcal{C}_C$	(29, 26, 21, 23)	(25, 24, 21, 22)	(27, 26, 21, 20)

¹ The baseline E_b/N_0 values vector $\boldsymbol{\eta}^*$ (in dB) for different channel conditions are specified in (19). The target FER is set to $p_e = 10^{-3}$.

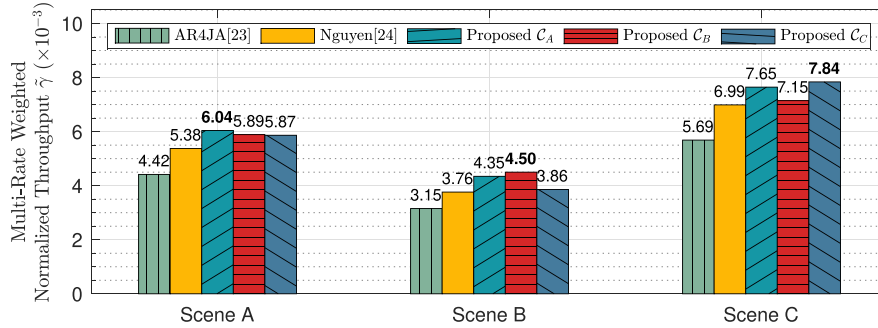


Fig. 7. Multi-rate weighted normalized throughput $\tilde{\gamma}$ comparison under the AWGN channel. Baseline E_b/N_0 values for code rates 1/2, 2/3, 3/4, and 4/5 are set as $\boldsymbol{\eta}_{\text{AWGN}}^* = (1.53, 2.12, 2.50, 2.80)$ dB.

iterations [20], [21] in simulation until the target FER is no longer met. The actual multi-rate weighted normalized throughput $\tilde{\gamma}$ can be calculated by substituting the simulation-derived required decoding iterations I_d^{sim} into (6).

In this work, the target FER is set to $p_e = 10^{-3}$. To evaluate the throughput performance, we adopt the baseline E_b/N_0 vector $\boldsymbol{\eta}^*$, defined as the lowest E_b/N_0 values at which Nguyen's codes achieve $p_e = 10^{-3}$ under a maximum of 20 decoding iterations for the four code rates. According to Figs. 4, 5 and 6, the baseline vectors $\boldsymbol{\eta}^*$ s (in dB) are set as

$$\begin{aligned}\boldsymbol{\eta}_{\text{AWGN}}^* &= (1.53, 2.12, 2.50, 2.80), \\ \boldsymbol{\eta}_{\text{Rician}, K=5 \text{ dB}}^* &= (3.33, 4.30, 5.03, 5.62), \\ \boldsymbol{\eta}_{\text{Doppler}, \Delta f_d T_s=1 \times 10^{-4}}^* &= (1.92, 2.50, 2.85, 3.17).\end{aligned}\quad (19)$$

The required decoding iterations ($I_1^{\text{sim}}, I_2^{\text{sim}}, I_3^{\text{sim}}, I_4^{\text{sim}}$) across the considered channels are summarized in Table IV. The simulated iterations I_d^{sim} deviate from the estimated value I_d predicted by PEXIT analysis. This is because I_d is derived using the PEXIT algorithm based on the decoding threshold, which only reflects the minimum E_b/N_0 at which successful decoding becomes theoretically possible. In contrast, I_d^{sim} is obtained under a practical performance constraint, where the FER is fixed at 10^{-3} . Nevertheless, the simulation results exhibit the same multi-rate weighted trend as the theoretical predictions across all channel conditions. Specifically, $\mathcal{C}_A$ achieves performance comparable to the benchmark Nguyen's code at all rates, with

an additional iteration count of no more than 4. $\mathcal{C}_B$ demonstrates competitive or even superior performance at low code rates (1/2 and 2/3), but requires notably more iterations at higher rates. For instance, in the AWGN channel, it needs 8 additional iterations over Nguyen's code at the 4/5 rate. In contrast, $\mathcal{C}_C$ requires fewer iterations at code rates 3/4 and 4/5, while its performance at low rates is less competitive.

Based on the required iterations I_d^{sim} and the resource estimates E_{total} , the multi-rate weighted normalized throughput $\tilde{\gamma}$ values are calculated and compared in Figs. 7, 8 and 9, corresponding to the AWGN, partial Rician fading, Doppler-affected channel, respectively. It can be observed that the numerical results obtained from simulations are consistent with the PEXIT theoretical analyses in Section IV-B. In general, the proposed codes $\mathcal{C}_A$, $\mathcal{C}_B$, $\mathcal{C}_C$ achieve the highest multi-rate weighted normalized throughput in Scene A, Scene B, and Scene C, respectively, which aligns with the intended design objectives. Taking the partial Rician fading channel as an example, $\mathcal{C}_A$ achieves a $\tilde{\gamma}$ improvement of 58.67% over the AR4JA code and 15.61% over Nguyen's code in Scene A. In Scene B, $\mathcal{C}_B$ provides gains of 58.10% and 19.41% over the two benchmarks, respectively. In Scene C, $\mathcal{C}_C$ outperforms the AR4JA code and Nguyen's code by 61.75% and 16.17%, respectively. Notably, even when applying a non-targeted code, e.g., $\mathcal{C}_B$ in Scene A, considerable multi-rate weighted normalized throughput improvements are still observed. For instance, under the partial Rician channel, $\mathcal{C}_B$ in Scene A achieves gains of 51.02% over the AR4JA code and 10.04% over Nguyen's code.

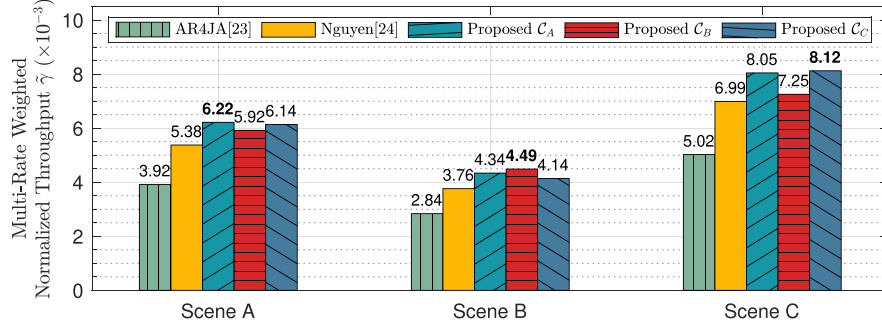


Fig. 8. Multi-rate weighted normalized throughput $\tilde{\gamma}$ comparison under the partial Rician fading channel with $K = 5$ dB. Baseline E_b/N_0 values for code rates 1/2, 2/3, 3/4, and 4/5 are set as $\eta_{\text{Rician}, K=5 \text{ dB}}^* = (3.33, 4.30, 5.03, 5.62)$ dB.

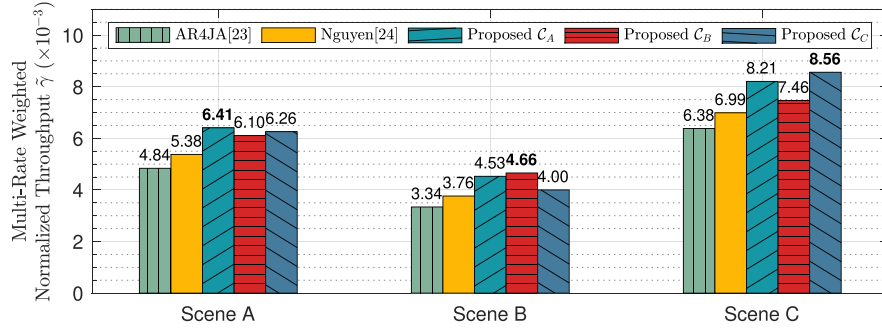


Fig. 9. Multi-rate weighted normalized throughput $\tilde{\gamma}$ comparison under the Doppler-affected channel with NFO $\Delta f_d T_s = 1 \times 10^{-4}$. Baseline E_b/N_0 values for code rates 1/2, 2/3, 3/4, and 4/5 are set as $\eta_{\text{Doppler}, \Delta f_d T_s = 1 \times 10^{-4}}^* = (1.92, 2.50, 2.85, 3.17)$ dB.

It is worth noting that the primary factors underlying the performance improvements over the two benchmarks differ in nature. Taking C_A as a representative example, the normalized throughput advantage over the AR4JA code is primarily attributed to a substantial reduction in the number of decoding iterations (averaging over 30%). In contrast, the improvement over Nguyen's code is mainly due to lower hardware resource consumption, with the total estimated resource E_{total} reduced by 19.4%. These comparative results further demonstrate the effectiveness of the proposed multi-rate weighted normalized throughput metric, which incorporates both resource usage and decoding iterations.

D. FER Comparison With the Same Multi-Rate Weighted Normalized Throughput

In practical communication systems, the required decoding throughput can be fulfilled either by adjusting the number of decoding iterations or by increasing the number of parallel decoder cores. When a specified throughput is achieved under fixed hardware constraints, throughput-optimized LDPC codes offer the advantage of allowing more decoding iterations, thereby enabling better error-correction performance. To demonstrate this, a representative operating point of C_A is selected from Fig. 7, where the decoding iteration configuration $(I_1^{\text{sim}}, I_2^{\text{sim}}, I_3^{\text{sim}}, I_4^{\text{sim}}) = (23, 20, 21, 23)$ yields a multi-rate weighted normalized throughput $\tilde{\gamma} = 6.14 \times 10^{-3}$ in Scene A.

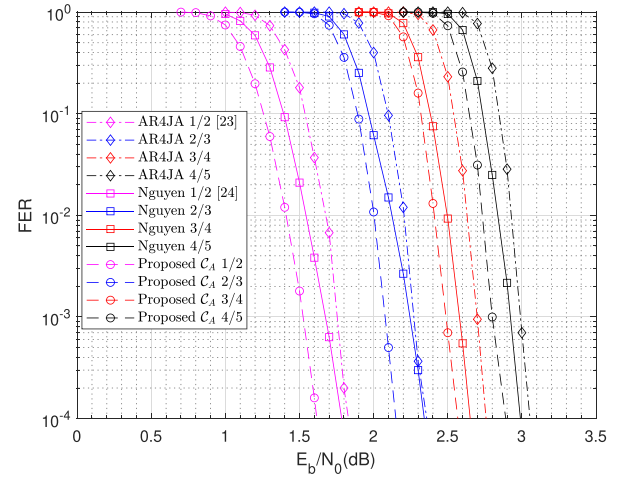


Fig. 10. FER performance comparison at the same multi-rate weighted normalized throughput levels. Operation point configurations are specified in Table V.

For fair comparison, the decoding iterations of the benchmark codes are set to (18, 18, 18, 17) for Nguyen's code and (21, 21, 21, 21) for the AR4JA code, achieving comparable throughput levels. The corresponding configurations are summarized in Table V.

The corresponding FER performance curves are shown in Fig. 10. At the same normalized throughput $\tilde{\gamma}$ level, the FER

TABLE V
OPERATING POINT CONFIGURATIONS AT THE SAME MULTI-RATE WEIGHTED
NORMALIZED THROUGHPUT LEVEL

Channel	Compared Codes	Decoding Iterations ($I_1^{\text{sim}}, I_2^{\text{sim}}, I_3^{\text{sim}}, I_4^{\text{sim}}$)	$\bar{\gamma}$ ($\times 10^{-3}$, Scene A)
AWGN	Proposed C_A	(23, 20, 21, 23)	6.14
	AR4JA Code [23]	(21, 21, 21, 21)	6.11
	Nguyen's Code [24]	(18, 18, 18, 17)	6.11

curves of the proposed throughput-optimized code exhibit a steeper slope in the waterfall region. In particular, C_A achieves at least 0.1 dB coding gain over Nguyen's code at a FER level of 10^{-3} , and around 0.2 dB gain over the AR4JA code across all code rates. These results confirm that the proposed throughput-optimized LDPC codes can achieve improved decoding performance under the same throughput and hardware resource constraints.

VI. CONCLUSION

This work proposes a new design method for constructing throughput-optimized multi-rate LDPC codes tailored to high-speed data transmission satellite communication with restricted decoder complexity. The study begins by analyzing key factors affecting LDPC decoder resource usage and decoding throughput from the perspective of code construction, leading to the introduction of the multi-rate weighted normalized throughput metric for multi-rate LDPC codes. Based on this metric, a constrained optimization problem is formally defined using the PEXIT algorithm, and an efficient optimization method based on the differential evolution algorithm is developed to solve it. The proposed design approach significantly reduces the number of non-zero elements in the parity-check matrix and minimizes decoding iterations across all code rates. In addition, different codes are optimized to match specific application scenes in which different levels of importance are assigned to the code rates. Simulation results under various channel conditions show that the proposed LDPC codes achieve lower resource consumption and fewer decoding iterations, ultimately resulting in substantial improvements in decoding throughput per unit of resource usage. These advantages are validated by comprehensive theoretical analyses and numerical simulations, confirming the effectiveness and adaptability of the proposed approach.

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